

Deepak Kumar
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Summary

Backend-specialized Software Engineer with strong foundations in Mathematics and Computing. Proven track record of building scalable MERN stack applications and high-performance Desktop Tools using Python/OpenCV. Experienced in designing secure authentication systems (JWT, OAuth 2.0) and real-time communication architectures. Seeking to contribute research-driven solutions to innovative engineering teams.

Education

Birla Institute of Technology, Mesra — May 2027 (Expected)
Integrated M.Sc. in Mathematics and Computing | CGPA: 8.58/10
Ranchi, Jharkhand

- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, DBMS.

Technical Skills

Languages: C++, Java, JavaScript (ES6+), Python, SQL

Backend & Database: Node.js, Express.js, REST APIs, JWT, MongoDB, MySQL

Frontend & Libraries: React.js, Tailwind CSS, Cheerio, Redux

DevOps & Tools: Git, GitHub Actions, Docker, Postman

Experience

Full Stack Developer (Freelance)

LooksLove Salon | Remote

Oct 2023 – Dec 2023

- Delivered a production-ready booking system with secure OTP-based login, scaling to 500+ active users and increasing digital appointment scheduling by 20%.
- Built a comprehensive Admin Dashboard (React/Node.js) for real-time inventory tracking and staff management, reducing daily administrative overhead by 30%.

Projects

1. LectureLens AI

Tech Stack: Python, OpenCV, NumPy, Tkinter

GitHub: github.com/deepak71kr

- Engineered a desktop video processing tool using OpenCV that removes 95% of visual obstructions from lectures via a custom Progressive Background Building algorithm.
- Optimized rendering efficiency by 40% using a Dual-Resolution strategy (processing low-res grayscale for motion analysis), generating 1080p PDF notes without quality loss.

2. Unlinked

Tech Stack: MERN Stack, Socket.io, JWT, Cloudinary

Live Demo: un-linked-7jqj.onrender.com

- Architected a professional networking platform featuring real-time messaging and live notifications using Socket.io, supporting simultaneous interactions for 20+ users.
- Implemented secure JWT Authentication and efficient REST APIs to manage complex social graph relationships (connections, likes) and optimized media storage via Cloudinary.

3. TnP Notifier

Tech Stack: Node.js, GitHub Actions, Cheerio, MongoDB

GitHub: github.com/deepak71kr/tnp-notifi

- Developed an automated placement alert system serving 200+ students, eliminating manual monitoring and saving approximately 15 hours of collective student time weekly.
- Orchestrated self-healing GitHub Actions CRON workflows with 99.9% uptime, ensuring zero missed notifications for critical job announcements.

Publications

OAuth 2.0 Security Analysis

Paper ID: 328 (Under Review)

- Authored a research paper analyzing authorization code flow vulnerabilities; proposed a token integrity protocol to mitigate CSRF and Replay Attacks.

Achievements

- **Problem Solving:** Solved 400+ DSA questions across various coding platforms (LeetCode, GFG).
- **IEEE Conference Presenter (2025):** Presented poster on "OAuth 2.0 in Healthcare" at NIT Jalandhar. [Drive Link : <https://drive.google.com/file/d/1cRYmYTcJq3OwGuaTcmfzEBdIgmWkc4Es/view?usp=sharing>]
- **Startup Founder (2022–2023):** Founded Earthen Pots, leading a team of 2 to manage production and D2C operations.
- **Volunteer (2022):** International Mathematics Conference, assisting in event organization and logistics.
- **NCC 'C' Certificate:** Demonstrated discipline, physical fitness, and leadership in annual training camps.